

very similar. Lemma 3.2 will be applied for self-similar measures μ , but we do not yet need to assume that μ is self-similar.

Lemma 3.2. *Let μ be a compactly supported Borel probability measure on $\mathbb{R}^k$ and let $f: \mathbb{R}^k \rightarrow \mathbb{R}^d$ be continuous. Consider the map $T: \mathbb{R}^k \rightarrow \mathbb{R}^{k+d}$, $\mathbf{x} \mapsto (\mathbf{x}, f(\mathbf{x}))$ and the image measure μ_T . Then for all $\boldsymbol{\xi} \in \mathbb{R}^d$, letting $\mathbf{0}$ be the zero vector with k entries,*

$$\widehat{\mu_T}(\mathbf{0}, \boldsymbol{\xi}) = \widehat{\mu_f}(\boldsymbol{\xi}).$$

Proof. This is simply a matter of unpacking the definition of the Fourier transform from (1.1). $\square$

Now μ be a self-similar measure on $\mathbb{R}^k$. Without loss of generality, we can assume that the support of μ is contained in a unit cube. Let Λ be an IFS for μ . For each $\omega \in \Lambda^*$ (the space of finite-length sequences with symbols in Λ), following the composition of maps $\omega_0\omega_1\cdots$, there is a least $l \geq 0$ such that the contraction ratio corresponding to $\omega_0^l = \omega_0\cdots\omega_l$ is at most $1/2^n$. We write $l = l_{\omega,n}$. Notice that the finite collection of paths

$$\{\omega_0^{l_{\omega,n}}\}_{\omega \in \Lambda}$$

corresponds to a covering of $\text{supp}(\mu)$. Such a covering uses similar copies of $\text{supp}(\mu)$ of sizes at most $1/2^n$ and at least $\rho_m 2^{-n}$ where ρ_m is smallest contraction ratio for μ . Let $\mu_{\omega,n}$ be the corresponding similar copy of μ with probability weight $|\mu_{\omega,n}|$. Clearly, each such branch $\text{supp}(\mu_{\omega,n})$ intersects at least one and at most 2^k many dyadic cubes in $\mathcal{D}_n$. We choose only one such dyadic cube and associate it with $\mu_{\omega,n}$. As a result, for each dyadic cube $D_n \in \mathcal{D}_n$, there is a collection of similar copies $\mu_{\omega,n}$ that are associated with this cube. We write $\mu_{\omega,n} \sim D_n$ for this association. We will call such a collection of ω to be Λ_{D_n} . We then obtain the following decomposition of Λ

$$\{\Lambda_{D_n}\}_{D_n \in \mathcal{D}_n}$$

which induces a decomposition (or disintegration) of μ . For each $\omega \in \Lambda_{D_n}$, the copy $\mu_{\omega,n}$ is supported in $3D_n$, the tripling of D_n with the same centre.

Given a C^2 map $f: \mathbb{R}^k \rightarrow \mathbb{R}^d$, write $\Gamma_f = \{(\mathbf{x}, f(\mathbf{x})) : \mathbf{x} \in \mathbb{R}^k\} \subset \mathbb{R}^{k+d}$ for the graph of f . Let $T_{\omega,n}$ be the (k -dimensional) tangent plane of Γ_f at $\mathbf{y}_0 = (\mathbf{x}_0, f(\mathbf{x}_0))$ for $\mathbf{x}_0$ being the centre of D_n . Given $\mu_{\omega,n} \sim D_n \subset \mathbb{R}^k$ and $\boldsymbol{\xi} \in \mathbb{R}^{k+d}$ with first k coordinates 0, let $\boldsymbol{\xi}_{\omega,n} \in \mathbb{R}^k$ be such that $(\boldsymbol{\xi}_{\omega,n}, \mathbf{0})$ is the projection of $\boldsymbol{\xi}$ to $\mathbb{R}^k \times \mathbf{0} \subset \mathbb{R}^{k+d}$ along the direction that is orthogonal to $T_{\omega,n}$, i.e.

$$(3.1) \quad ((\boldsymbol{\xi}_{\omega,n}, \mathbf{0}) - \boldsymbol{\xi}) \perp T_{\omega,n}.$$

Note that in the following statement we do not yet assume that f has positive Gaussian curvature.

Lemma 3.3. *Let μ be a self-similar measure on $\mathbb{R}^k$, let $f: \mathbb{R}^k \rightarrow \mathbb{R}^d$ be C^2 , and let $T(\mathbf{x}) = (\mathbf{x}, f(\mathbf{x}))$. Then for sufficiently large $n \in \mathbb{N}$ and $\boldsymbol{\xi} \in \mathbb{R}^{k+d}$*